



316-221Metallurgy

Chapter 1

Arrangement of Atoms and Crystal Structure

The Arrangement of Atoms in Crystal Structure

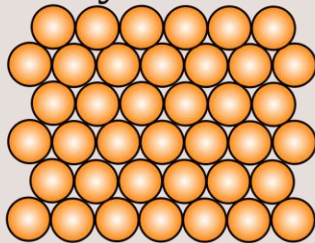
The arrangement of atoms in crystal structure is one of the most important aspects of materials science and engineering as many properties of materials depend on their crystal structures. The basic principles of many materials characterization techniques such as X-ray diffraction (XRD). Therefore, understanding the basics of crystal structures is of paramount importance.

In gases, atoms have no order.

If atoms bonded to each other but there is no repeating pattern (short range order) . e.g. water, glasses (**AMORPHOUS - non-crystalline**)

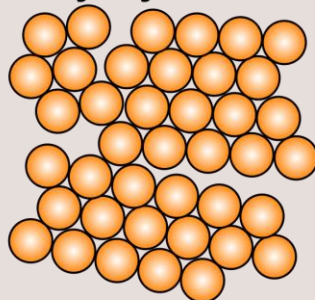
If atoms bonded together in a regular 3-D pattern they form a **CRYSTAL** - long range order - like wall paper pattern or brick wall.

Crystalline

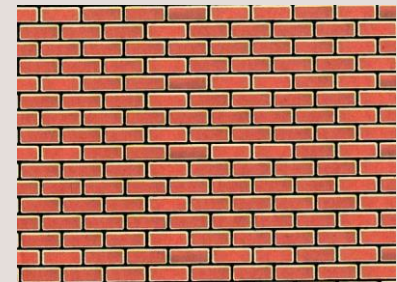
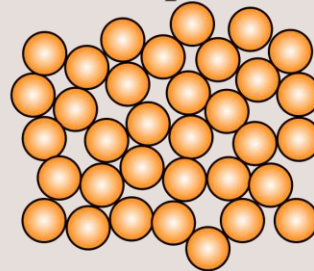


22/06/2023

Polycrystalline



Amorphous



The Arrangement of Atoms in Crystal Structure

For solid materials:

Most **METALS** (>99%) are **Crystalline**.

CERAMICS are **Crystalline** except for **GLASSES** which are **Amorphous**.

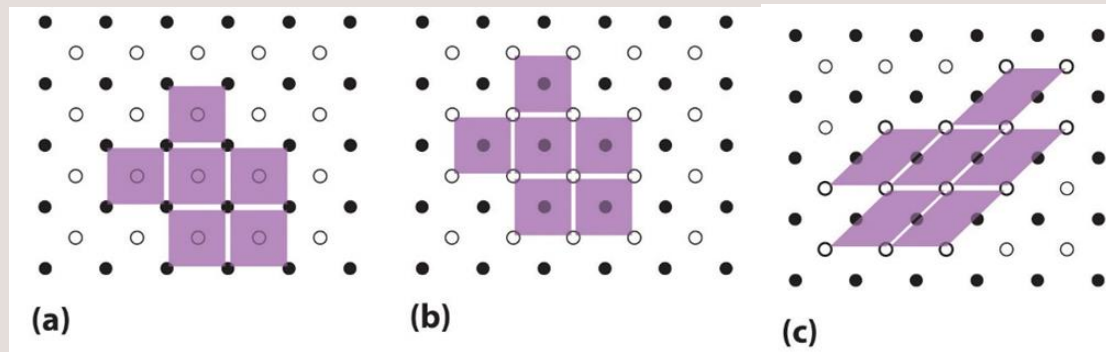
POLYMERS (plastics) tend to be:

either **Amorphous** or a mixture of **Crystalline** + **Amorphous**.
(known as Semi-crystalline)

The Arrangement of Atoms in Crystal Structure

Crystalline solid consists of repeating patterns of its components in three dimensions (**a crystal lattice**), we can represent the entire crystal by drawing the structure of the smallest identical units that, when stacked together, form the crystal. This basic repeating unit is called a **unit cell**.

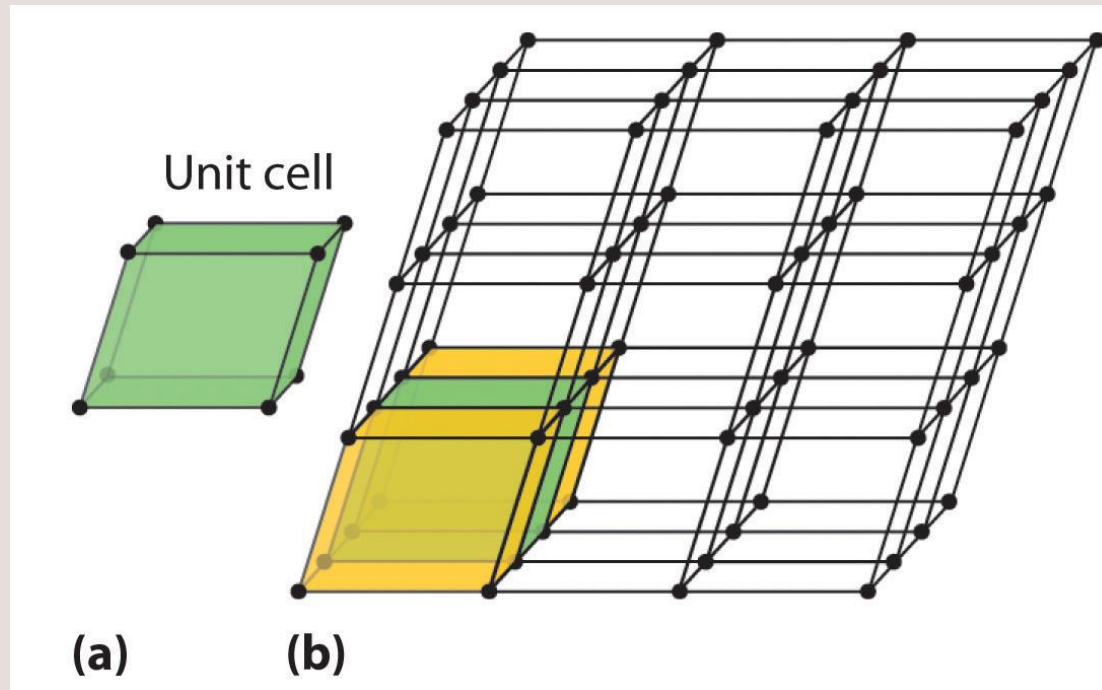
Unit cells are easiest to visualize in two dimensions. In many cases, more than one unit cell can be used to represent a given structure, in figure shows **a two-dimensional (2D) crystal lattice**.



(a–c) Three two-dimensional lattices illustrate the possible choices of the unit cell.

The Arrangement of Atoms in Crystal Structure

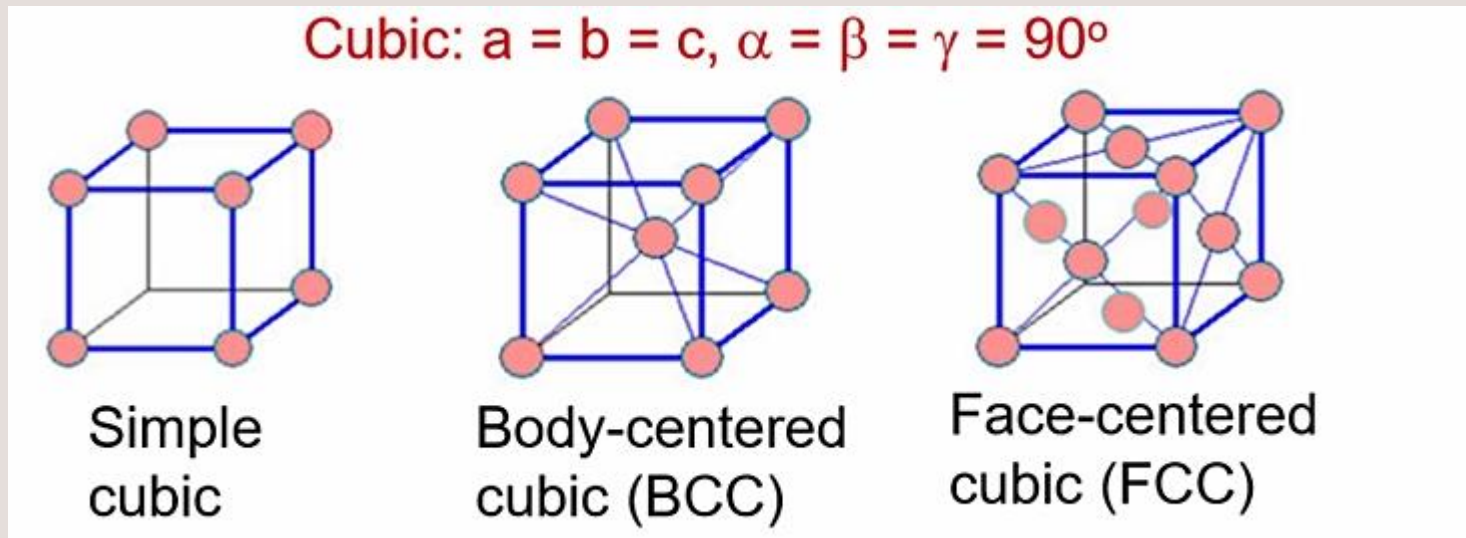
The concept of unit cells is extended to a three-dimensional lattice in the schematic drawing in figure below **"Unit Cells in Three Dimensions (3D)"**.



These images show (a) a three-dimensional unit cell and (b) the resulting regular three-dimensional lattice.

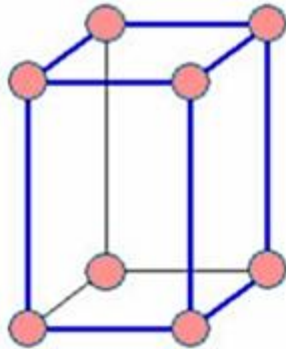
Bravais Lattice in Three dimension-Space lattice

The unit vectors a , b and c are called lattice parameters. Based on their length equality or inequality and their orientation (the angles between them, α , β and γ) a total of **7 crystal systems** can be defined. With the centering (**face, base and body centering**) added to these, **14 kinds** of 3D lattices, known as **Bravais lattices**, can be generated.

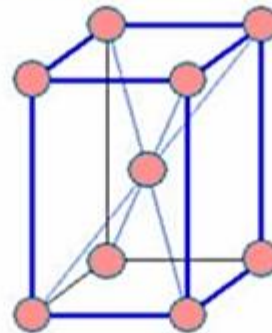


Bravais Lattice in Three dimension-Space lattice

Tetragonal: $a = b \neq c, \alpha = \beta = \gamma = 90^\circ$

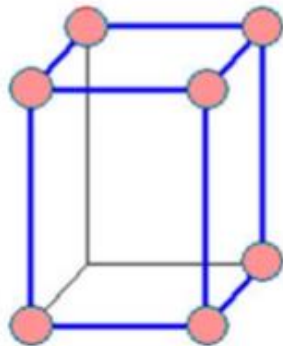


Simple
Tetragonal

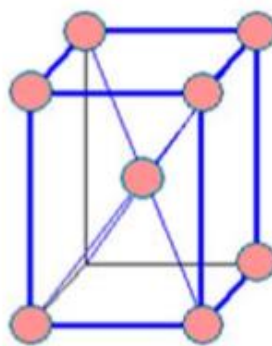


Body-centered
Tetragonal (BCT)

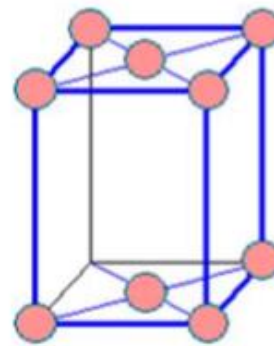
Orthorhombic: $a \neq b \neq c, \alpha = \beta = \gamma = 90^\circ$



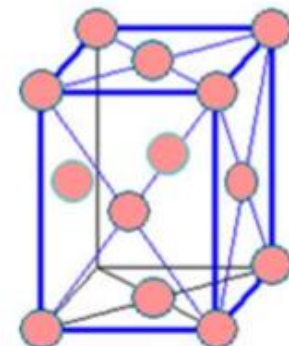
Simple



Body-centered



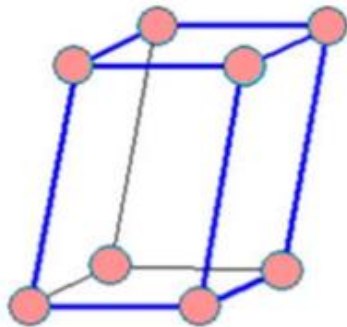
Base-centered



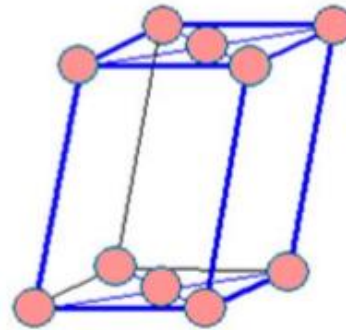
Face-centered

Bravais Lattice in Three dimension-Space lattice

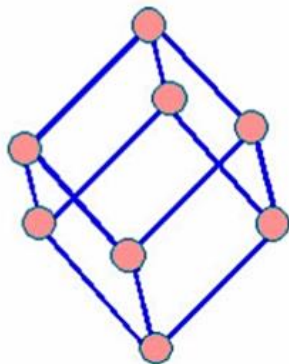
Monoclinic: $a \neq b \neq c$, $\alpha = \gamma = 90^\circ \neq \beta$



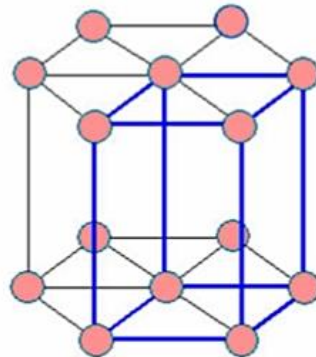
Simple
monoclinic



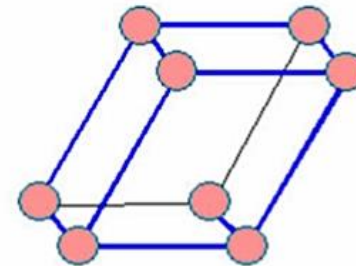
Base-centered
monoclinic



Rhombohedral
 $a = b = c$
 $\alpha = \beta = \gamma \neq 90^\circ$



Hexagonal
 $a = b \neq c$
 $\alpha = \beta = 90^\circ \neq \gamma = 120^\circ$



Triclinic
 $a \neq b \neq c$
 $\alpha \neq \beta \neq \gamma \neq 90^\circ$

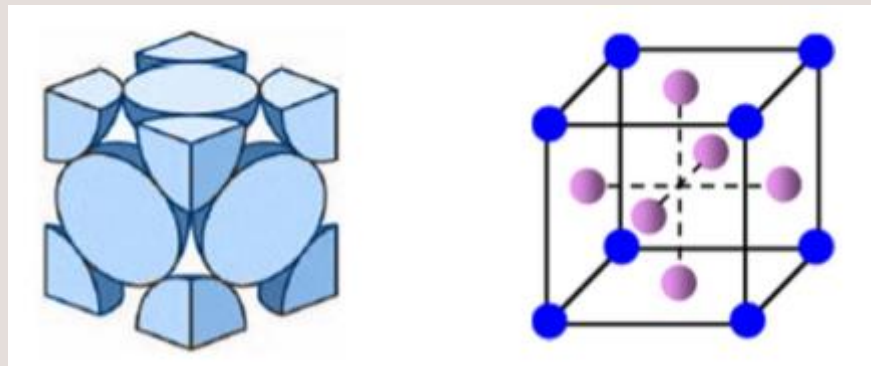
Bravais Lattice in Three dimension-Space lattice

No.	Crystal System	Axes	Angles	Examples
1.	Cubic	$a = b = c$	$\alpha = \beta = \gamma = 90^\circ$	Fe, Cu, NaCl, NaBr, Diamond
2.	Tetragonal	$a = b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	Sn, SnO ₂ , MnO ₂ , NH ₄ Br
3.	Orthorhombic	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	Iodine, KNO ₃ , Rhombic sulphur
4.	Rhombohedral	$a = b = c$	$\alpha = \beta = \gamma \neq 90^\circ < 120^\circ$	Bi, Al ₂ O ₃ , NaNO ₃ , KNO ₃
5.	Hexagonal	$a = b \neq c$	$\alpha = \beta = 90^\circ, \gamma = 120^\circ$	Graphite, ZnO, CdS
6.	Monoclinic	$a \neq b \neq c$	$\alpha = \gamma = 90^\circ, \beta > 90^\circ$	Sugar, Sulphur, Borax
7.	Triclinic	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90^\circ$	H ₃ BO ₃ , K ₂ Cr ₂ O ₇ , CuSO ₄ . 5H ₂ O

Metallic Crystal Structures and Atomic Packing Factor (APF)

Coordination number (CN)

- Coordination number is the number of nearest neighbor to a particular atom in the crystal



In the FCC lattice each atom is in contact with 12 neighbor atoms.

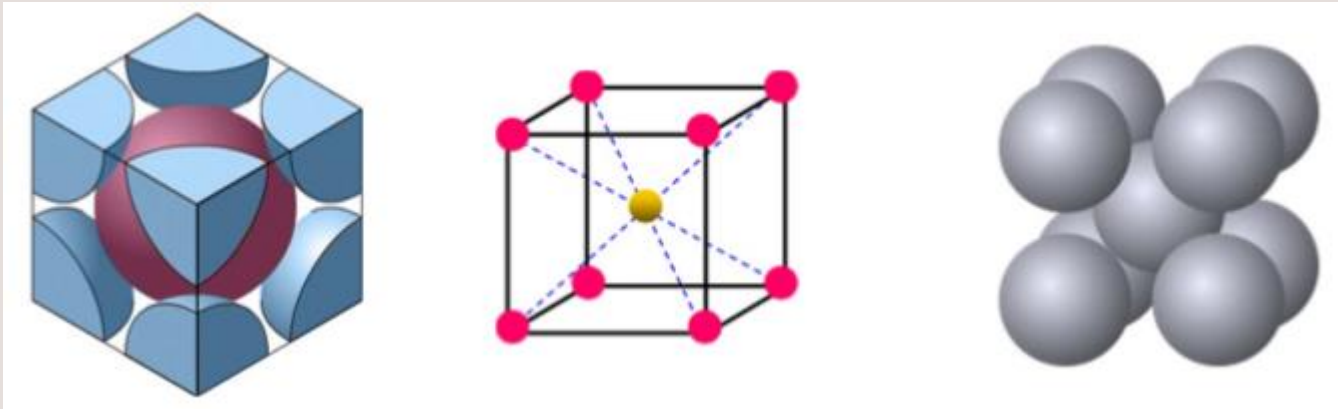
FCC coordination number (CN) = 12

For example, the face centered atom in the front face is in contact with four corner atoms and four other face-centered atoms behind it (two sides, top and bottom) and is also touching four face-centered atoms of the unit cell in front of it.

Metallic Crystal Structures and Atomic Packing Factor (APF)

Coordination number (CN)

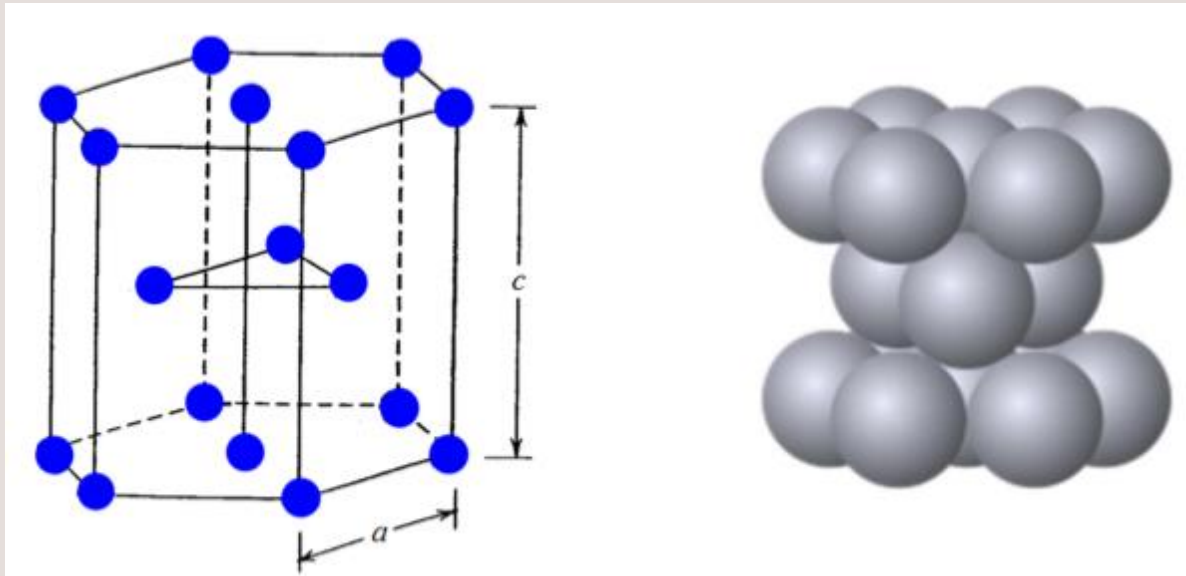
- The coordination number of BCC crystal is **8**



The body centered atom is in contact with all the eight corner atoms. Each corner atom is shared by eight unit cells and hence, each of these atoms is in touch with eight body centered atoms

Metallic Crystal Structures and Atomic Packing Factor (APF)

Coordination number (CN)



In Hexagonal lattice **CN = 12**. The center atom of the top face is in touch with six corner atoms, three atoms of the mid layer and other three atoms of the mid layer of the unit cell above it.

Metallic Crystal Structures and Atomic Packing Factor (APF)

Atomic packing factor (APF)

Atomic packing factor or packing efficiency indicates how closely atoms are packed in a unit cell and is given by the ratio of **volume of atoms in the unit cell** and **volume of the unit cell**

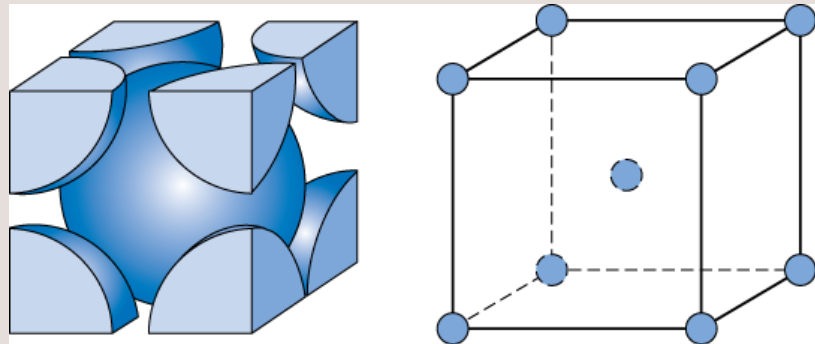
$$\text{APF} = \text{Atomic volume in unit cell} / \text{Unit cell volume}$$

Metallic Crystal Structures and Atomic Packing Factor (APF)

BCC lattice:

Unit Cell **BCC** type is an atom at the centre of the cube

($a = b = c$, $\alpha = \beta = \gamma = 90^\circ$)



Can be calculated of atom number per unit cell of **BCC**

Central atom of unit cell = 1

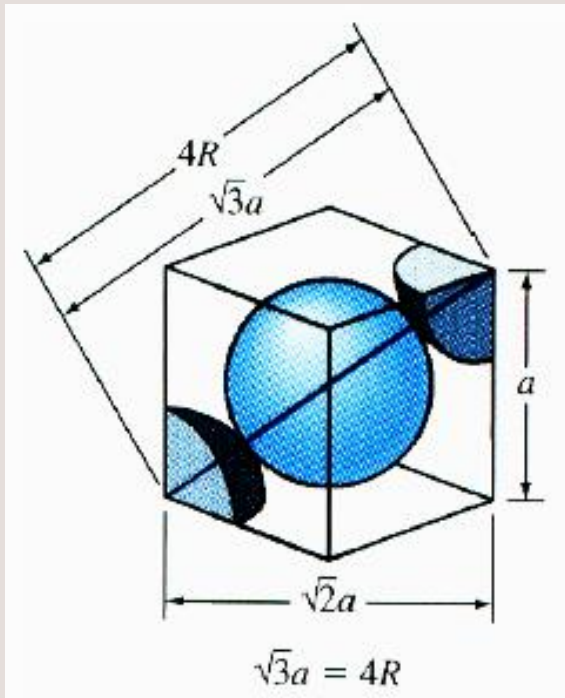
Corner atom at 1 corner = $1/8$

Corner atom at 8 corners = $(1/8) \times 8 = 1$

Total atom = $1 + 1 = 2$

Metallic Crystal Structures and Atomic Packing Factor (APF)

The relationship between a and R can be shown: $4R = \sqrt{3}a$



Thus, $a = \frac{4R}{\sqrt{3}}$

Unit cell volume (V_{volume}) $a = a^3$
 $= \left(\frac{4R}{\sqrt{3}}\right)^3$

Atomic volume (V_{atom}) $= \frac{4}{3} \pi R^3$

Because, BCC is 2 atoms $= 2 \left(\frac{4}{3} \pi R^3\right)$

From, $\text{APF} = \text{Atomic volume} / \text{Unit cell volume} = \frac{2 \left(\frac{4}{3} \pi R^3\right)}{\left(\frac{4R}{\sqrt{3}}\right)^3} = 0.68$

Thus, the area in 1 unit of BCC has 0.68 unit of atom and 0.32 unit of void

Metallic Crystal Structures and Atomic Packing Factor (APF)

Table show lattice constants and atomic radius of metal in **BCC** structure at temperature (20°C)

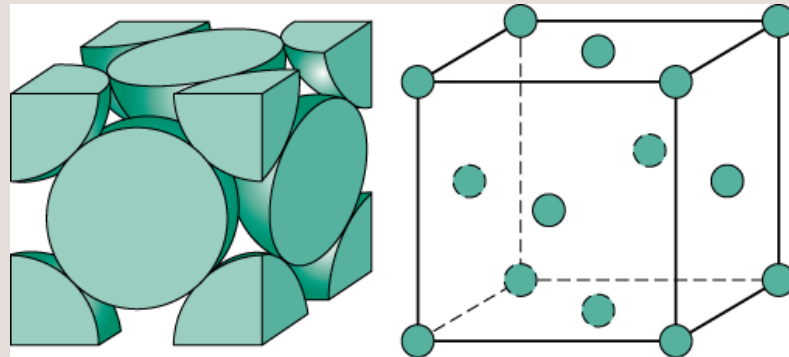
Metal	Lattice constant	Atomic radius
	a, (nm.)	R, (nm.)
Chromium	0.289	0.125
Iron	0.287	0.124
Molybdenum	0.315	0.136
Potassium	0.533	0.231
Sodium	0.429	0.186
Tantalum	0.330	0.143
Tungsten	0.316	0.137
Vanadium	0.304	0.132

Metallic Crystal Structures and Atomic Packing Factor (APF)

FCC lattice:

Unit Cell **FCC** type is an atom at 8 corners of the cube

$$(a = b = c, \alpha = \beta = \gamma = 90^\circ)$$



Can be calculated of atom number per unit cell of **FCC**

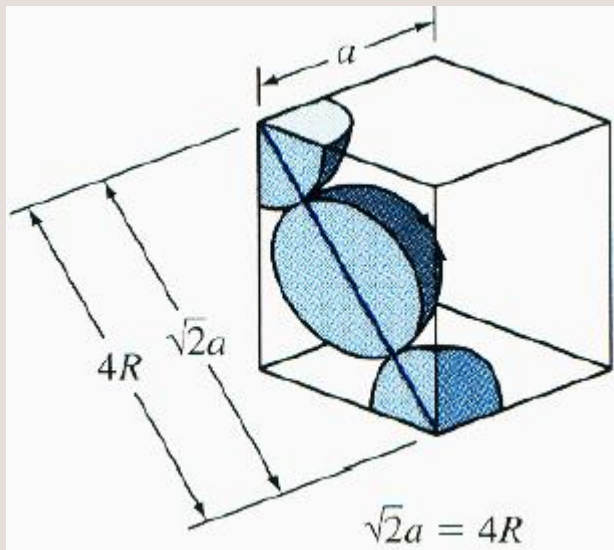
$$\text{Corner atom at 8 corners} = (1/8) \times 8 = 1$$

$$\text{Central atom at 6 surface} = (1/2) \times 6 = 3$$

$$\text{Total atom} = 1 + 3 = 4$$

Metallic Crystal Structures and Atomic Packing Factor (APF)

FCC crystal structure has 4 atoms contained per unit cell



The relationship of radius (R) to the lattice parameter (a) is $4R = \sqrt{2}a$

Thus,
$$a = \frac{4R}{\sqrt{2}}$$

Atomic Packing Factor (APF) in FCC = 0.74

APF in FCC > BCC

Thus, all atoms in FCC has dense arrangement than that in the BCC.

Metallic Crystal Structures and Atomic Packing Factor (APF)

Table show lattice constants and atomic radius of metal in **FCC** structure at temperature (20°C)

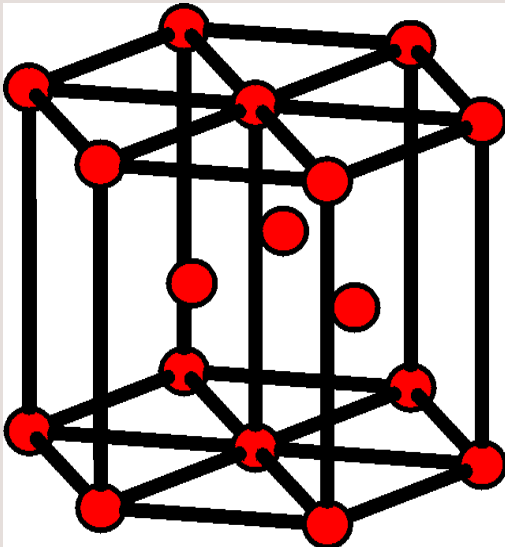
Metal	Lattice constant	Atomic radius
	a, (nm.)	R, (nm.)
Aluminum	0.405	0.143
Copper	0.3615	0.128
Gold	0.408	0.144
Lead	0.495	0.175
Nickel	0.352	0.125
Platinum	0.393	0.139
Silver	0.409	0.144

Metallic Crystal Structures and Atomic Packing Factor (APF)

HCP is a six faces $(a = b \neq c, \alpha = \beta = 90^\circ, \gamma = 120^\circ)$

Atomic Packing Factor (APF) in **HCP** = 0.74

HCP crystal structure has 6 atoms contained per unit cell



1 corner atom at 6 corners	=	1/6
Total corner atom 12 corners	=	$(1/6) \times 12 = 2$
Between upper and lower plane	=	3
At upper and lower plane	=	$(1/2) \times 2 = 1$
Total atom	=	$2 + 3 + 1 = 6$

Metallic Crystal Structures and Atomic Packing Factor (APF)

Table show lattice constants, atomic radius and c/a ratio of metal in **HCP** structure at temperature (20°C)

Metal	Lattice constant		Atomic radius R, (nm.)	c/a ratio	% deviation from ideality
	a, (nm.)	a, (nm.)			
Cadmium	0.2973	0.5618	0.149	1.890	+15.7
Zinc	0.2665	0.4947	0.133	1.856	+13.6
Ideal HCP				1.633	0
Magnesium	0.3209	0.5209	0.160	1.623	-0.66
Cobalt	0.2507	0.4069	0.125	1.623	-0.66
Zirconium	0.3231	0.5148	0.160	1.593	-2.45
Titanium	0.2950	0.4683	0.147	1.587	-2.81
Beryllium	0.2286	0.3584	0.113	1.568	-3.98

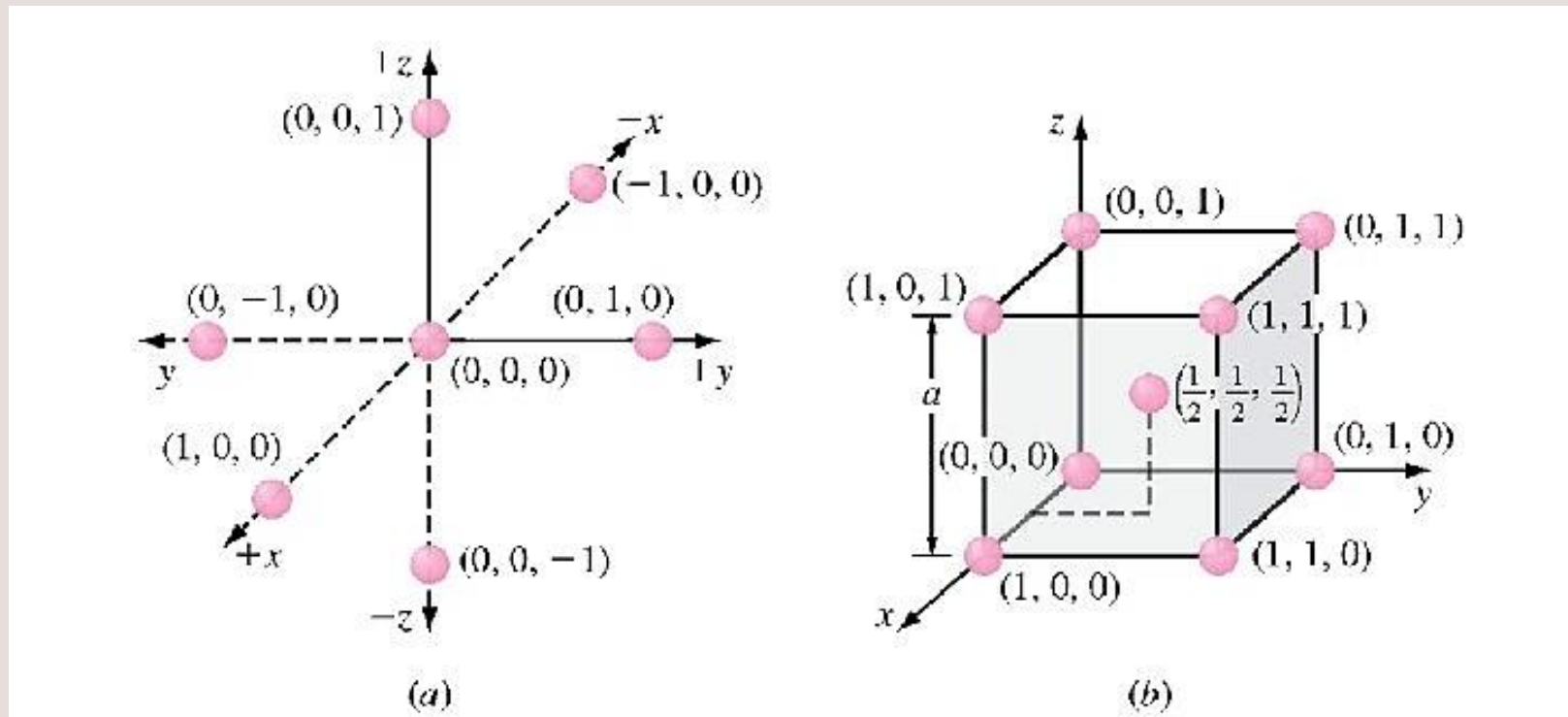
Metallic Crystal Structures and Atomic Packing Factor (APF)

Structure	a Vs R	Atom/cell	CN	APF	Typical metal
SC	$a = 2R$	1	6	0.52	None
BCC	$a = 4R / \sqrt{3}$	2	8	0.68	Fe, Ti, W, Mo, K, Na, V, Cr, Zr
FCC	$a = 4R / \sqrt{2}$	4	12	0.74	Fe, Cu, Al, Au, Ag, Pb, Ni, Pt
HCP	$a = 2R$ $c = 1.63a$	6	12	0.74	Ti, Mg, Zn, Be, Co, Zr, Cd

Lattice Points in Cubic Unit Cell

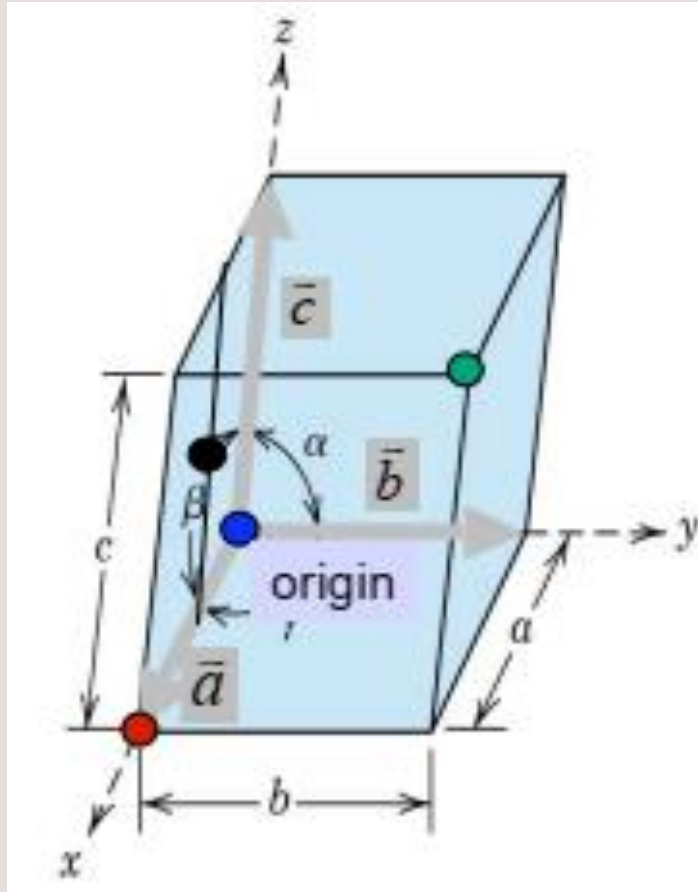
Miller-indices - A shorthand notation to describe certain crystallographic directions and planes in a material.

We start with the coordinates of lattice points in order to define the Miller indices of lattice directions



Lattice Points in Cubic Unit Cell

How to define Miller indices for point within a unit cell?



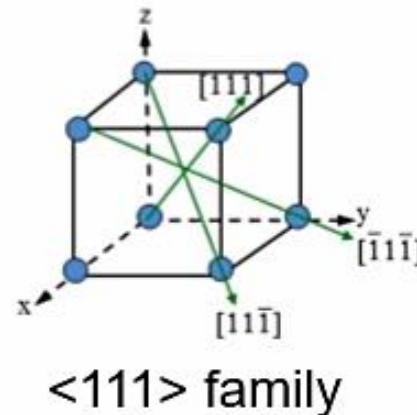
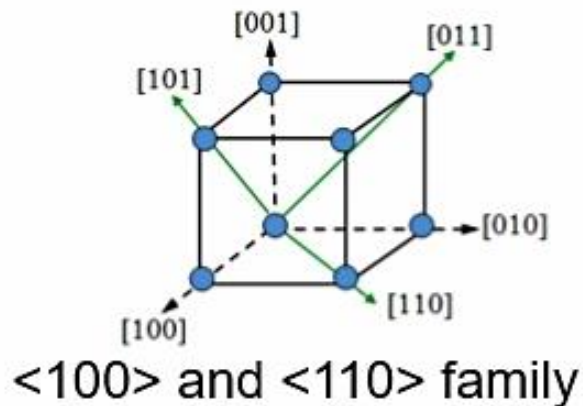
	pt. coord.		
pt.	$x(a)$	$y(b)$	$z(c)$
	0	0	0
	1	0	0
	1	1	1
	$1/2$	0	$1/2$

Crystallographic direction

The directions in a crystal are given by specifying the coordinates (x, y, z) of a point on a vector (r_{xyz}) passing through the origin. It is indicated as [xyz].

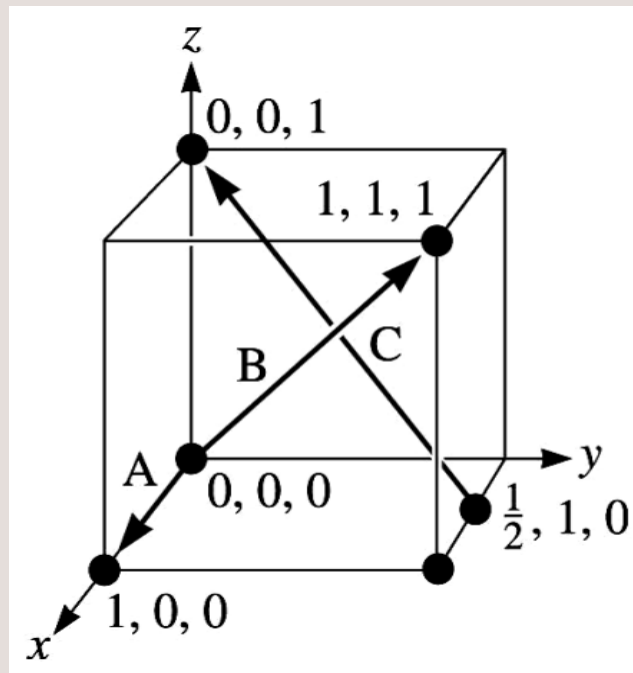
For example, the direction $[110]$ lies on a vector r_{110} whose projection lengths on x and y axis are one unit.

Directions of a form or family like $[110]$, $[101]$, $[011]$ are written as $\langle 110 \rangle$



Crystallographic direction

Determining miller indices of A, B and C directions.



Direction A

1. Two points are 1, 0, 0, and 0, 0, 0
2. $1, 0, 0, - (0, 0, 0) = 1, 0, 0$
3. No fractions to clear or integers to reduce
4. $[100]$

Direction B

1. Two points are 1, 1, 1 and 0, 0, 0
2. $1, 1, 1, - (0, 0, 0) = 1, 1, 1$
3. No fractions to clear or integers to reduce
4. $[111]$

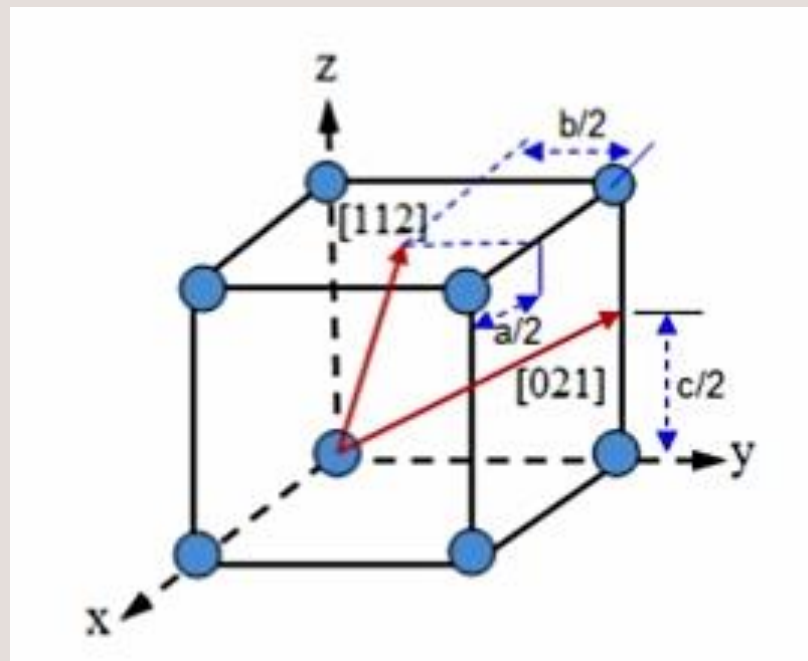
Direction C

1. Two points are 0, 0, 1 and $1/2, 1, 0$
2. $0, 0, 1 - (1/2, 1, 0) = -1/2, -1, 1$
3. $2 (-1/2, -1, 1) = -1, -2, 2$
4. $[-1 -2 2]$

Crystallographic direction

Drawing direction vectors in cubic unit cells of **[112]** and **[021]**.

- The line which passes through **xyz** will also pass through $\frac{1}{2}x \frac{1}{2}y \frac{1}{2}z$. Hence $[xyz]$, $[\frac{1}{2}x \frac{1}{2}y \frac{1}{2}z]$ are same and written as $[xyz]$
- Fractions are converted in to integers (as shown in the figure below) and reduced to lowest terms



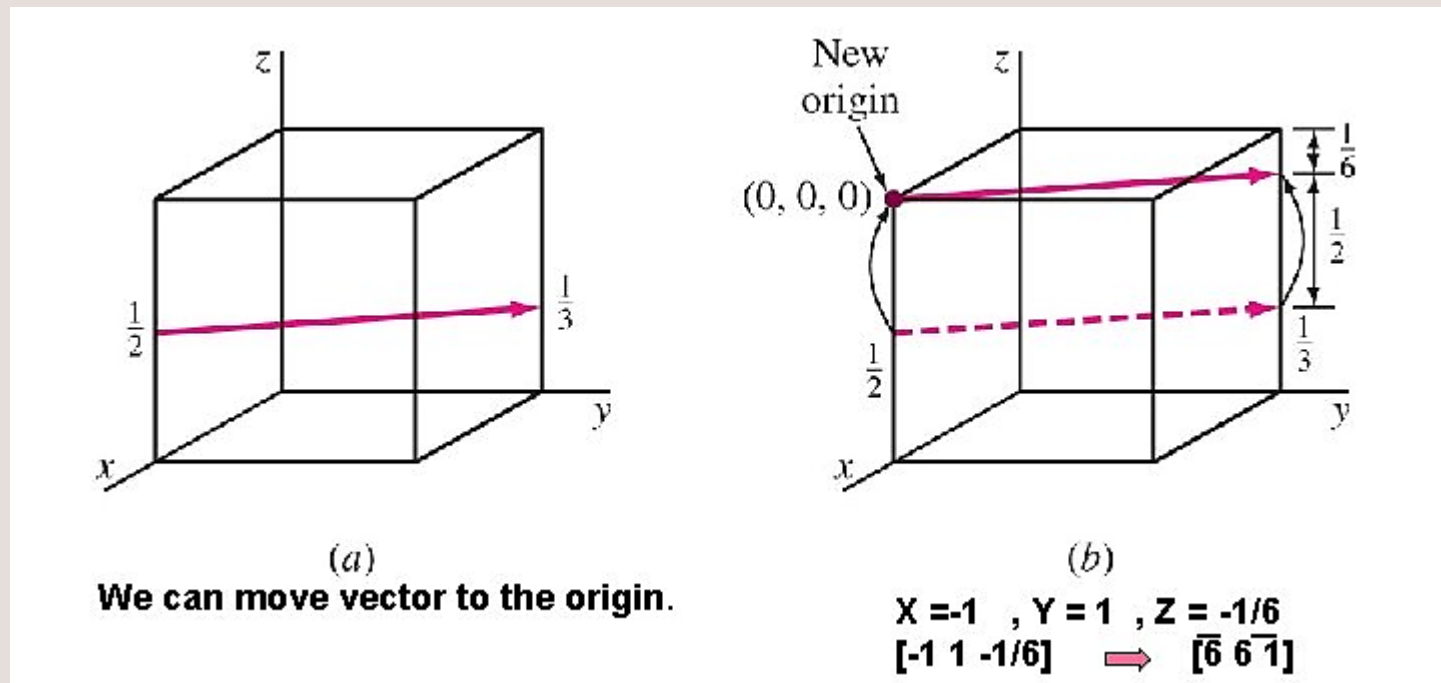
Crystallographic direction

Determining direction indices of cubic directions below:

Examples:

We can move vector to the origin. $X = -1$, $Y = 1$, $Z = -1/6$ $[-1 \ 1 \ -1/6]$

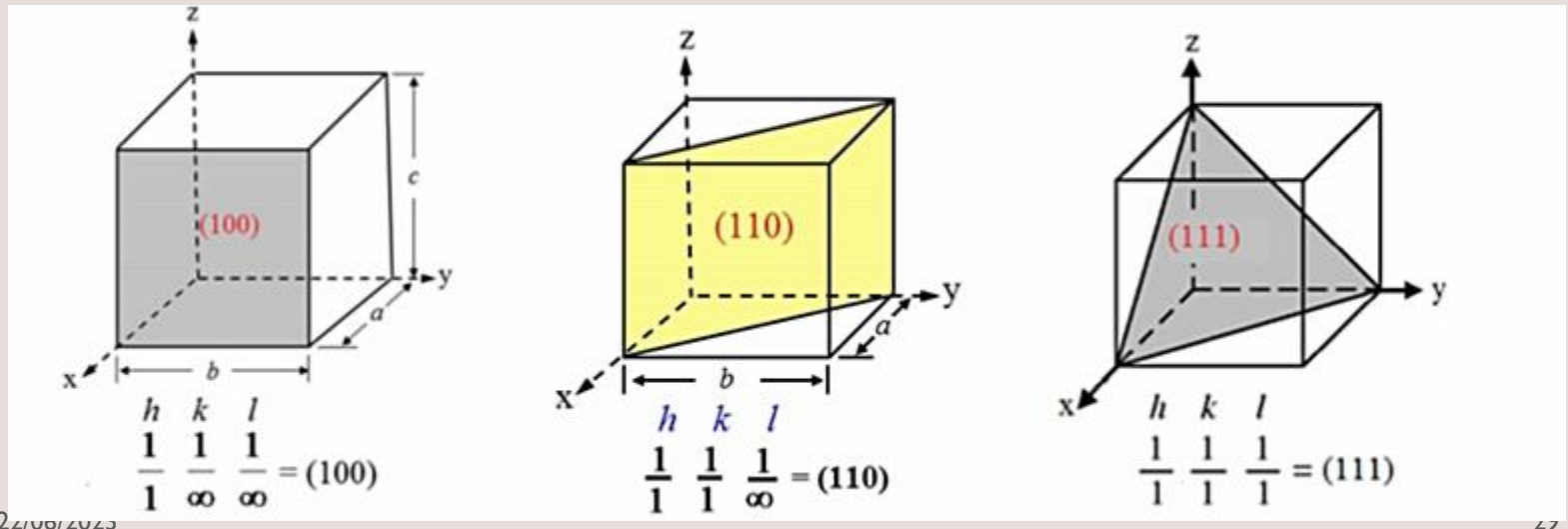
Direction indices $[-6 \ 6 \ -1]$



Miller Indices of crystallographic planes

Planes in a crystal are described by notations called **Miller indices**

- **Miller indices** of a plane, indicated by **$h\ k\ l$** , are given by the reciprocal of the intercepts of the plane on the three axis
- The plane, which intersects X axis at 1 (one lattice parameter) and is parallel to Y and Z axis, has Miller indices **$h = 1/1 = 1$, $k = 1/\infty = 0$, $l = 1/\infty = 0$** . It is written as $(hkl) = (100)$
- Miller indices of some other planes in the cubic system are shown in the figures

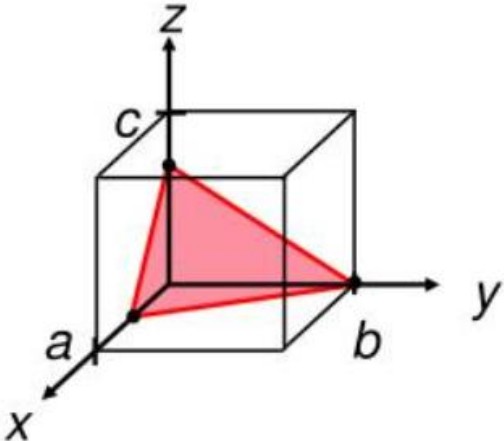


Miller Indices of crystallographic planes

To find the Miller Indices of a plane, follow these steps:

- Determine the intercepts of the plane along the crystal axis
- Take the reciprocals
- Clear fractions
- Reduce to lowest terms and enclose in brackets ()

<u>example</u>	<i>a</i>	<i>b</i>	<i>c</i>
1. Intercepts	$1/2$	1	$3/4$
2. Reciprocals	$1/1/2$	1/1	$1/3/4$
	2	1	$4/3$
3. Reduction	6	3	4
4. Miller Indices	(634)		



Miller Indices of crystallographic planes

Family of planes $\{hkl\}$

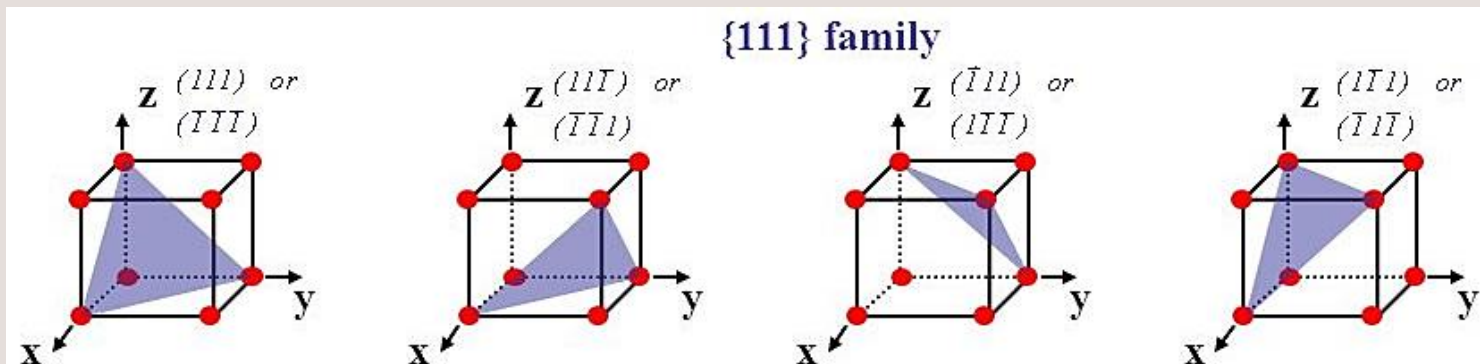
Planes having similar index are equivalent, e.g. faces of the cube (100), (010) and (001). This is termed as a family of planes and denoted as $\{100\}$ which includes all the (100) combinations including negative index.

Example: $\{111\}$

$= (111), (-111), (1-11), (11-1), (-1-1-1), (-1-11), (-11-1), (1-1-1)$

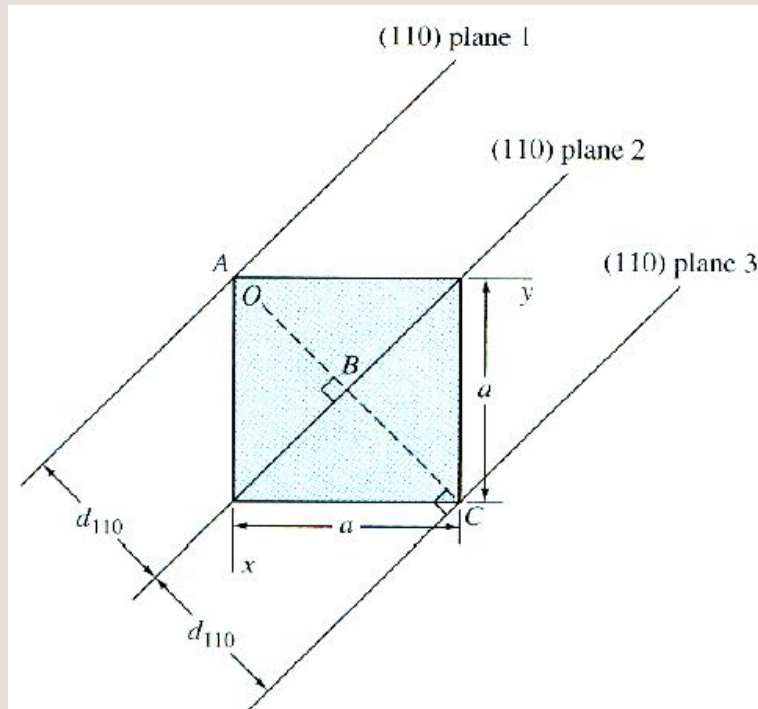
Example: $\{100\}$

$= (100), (010), (001), (-100), (0-10), (00-1)$



Crystallographic planes

The spacing between planes in a crystal is known as **interplanar spacing** and is denoted as d_{hkl}



$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

d_{hkl} is the interplanar spacing

a is the lattice constant

h, k, l are the miller indices of cubic planes

Example: Calculated the interplanar spacing between the (2 2 1) planes of a cubic lattice of length 450 pm.

$$\begin{aligned} d_{hkl} &= \frac{a}{\sqrt{h^2 + k^2 + l^2}} \\ &= \frac{450}{\sqrt{4 + 4 + 1}} = \frac{450}{\sqrt{9}} = 150 \text{ pm} \end{aligned}$$

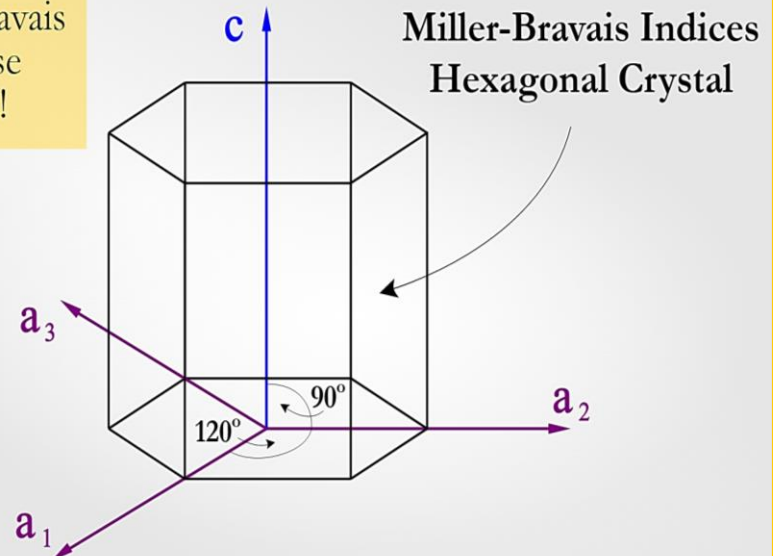
Crystallographic planes and directions in Hexagonal unit cells

How to read Hexagonal crystal directions and planes?

Miller-Bravais indices are a 4-axis coordinate system for 3-dimensional crystals, based on the unit cell. This coordinate system is based on the 3-axis Miller index, but with an extra axis which is used for hexagonal crystals. The system can indicate directions or planes, and are often written as $(hkil)$.

For Miller-Bravais indices, we need to label 4 axes in the hexagonal crystal. In the basal plane, we have 3 axes of equal length each separated by 120° , which we call a_1 , a_2 , and a_3 . Then there is the c axis, perpendicular to those three.

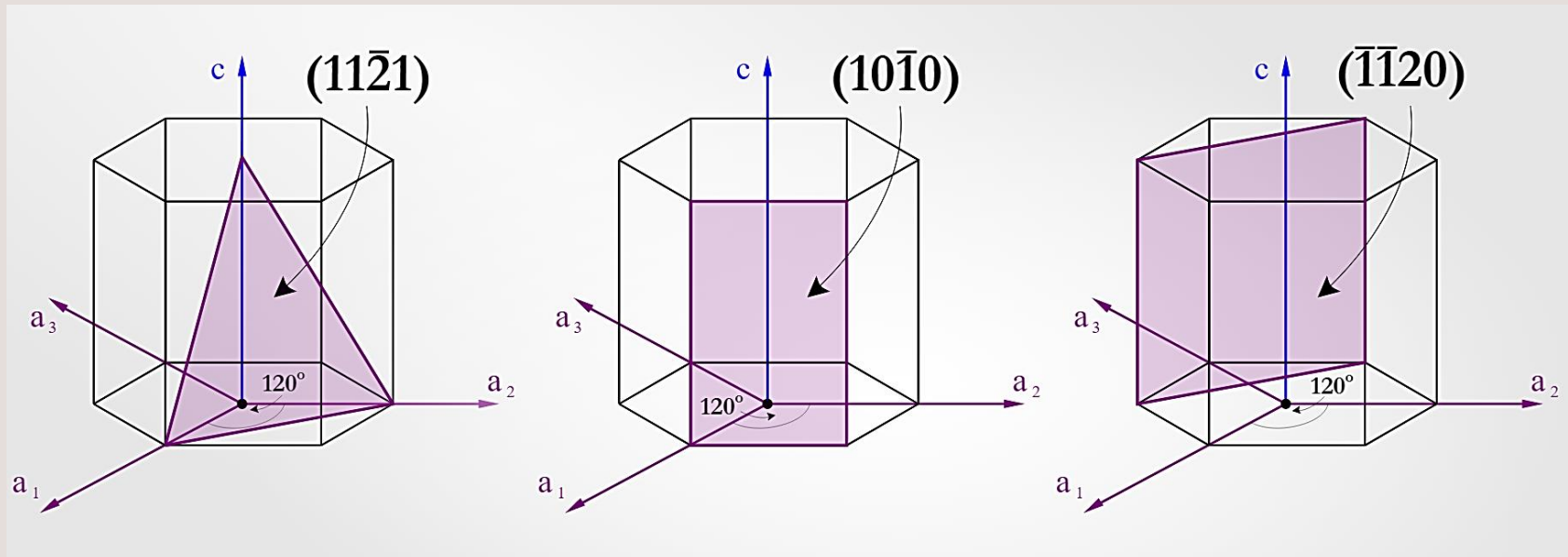
Miller-Bravais indices use four axes!



Crystallographic planes and directions in Hexagonal unit cells

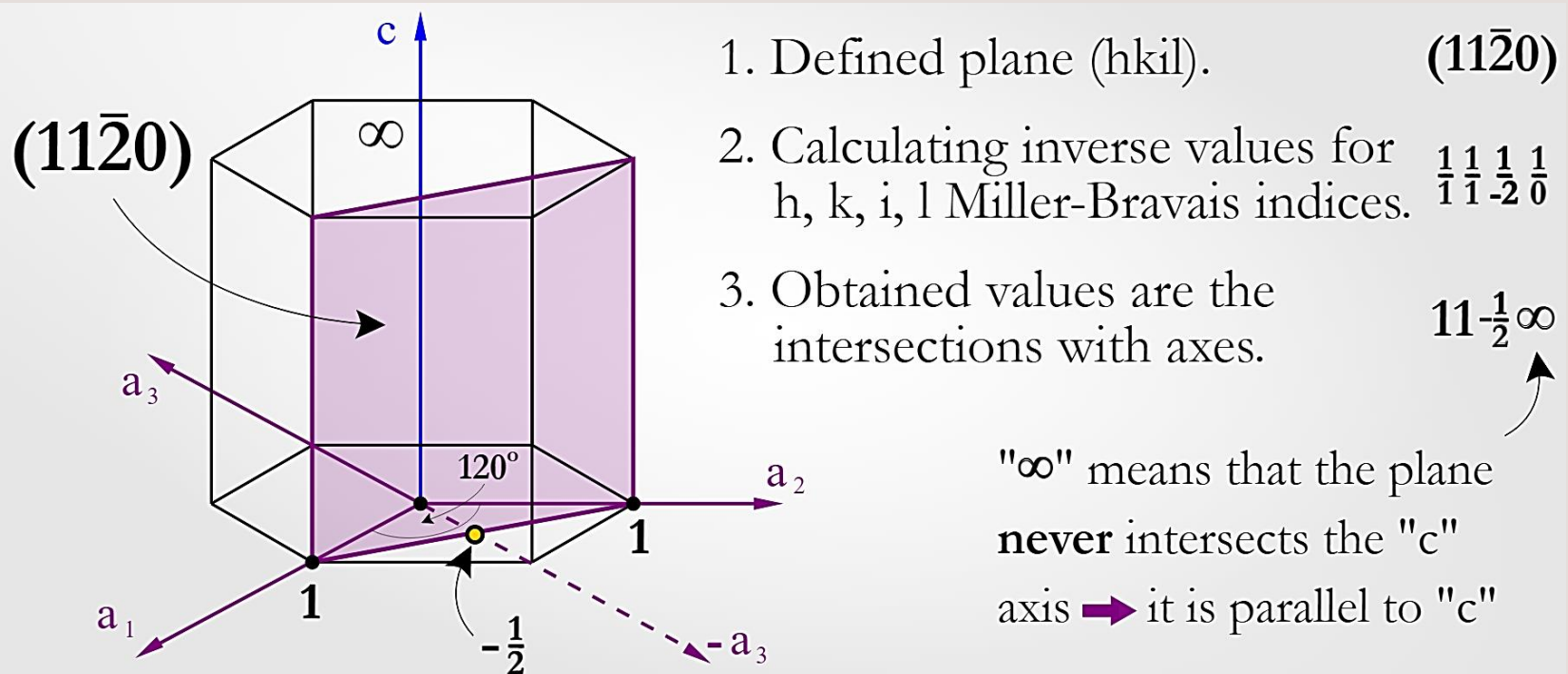
Planes in the 4-axis system ($hkil$) system are actually very similar to the 3-axis (hkl) system—that's because “h,” “k,” and “l” are the same in both systems. “i” is simply defined by the formula $h + k = -i$

Otherwise, you can use the regular procedure for finding planes in Miller indices. You don't even need an image to see that (111) becomes $(11\bar{2}1)$, (100) becomes $(10\bar{1}0)$, or $(\bar{1}10)$ becomes $(\bar{1}\bar{1}20)$



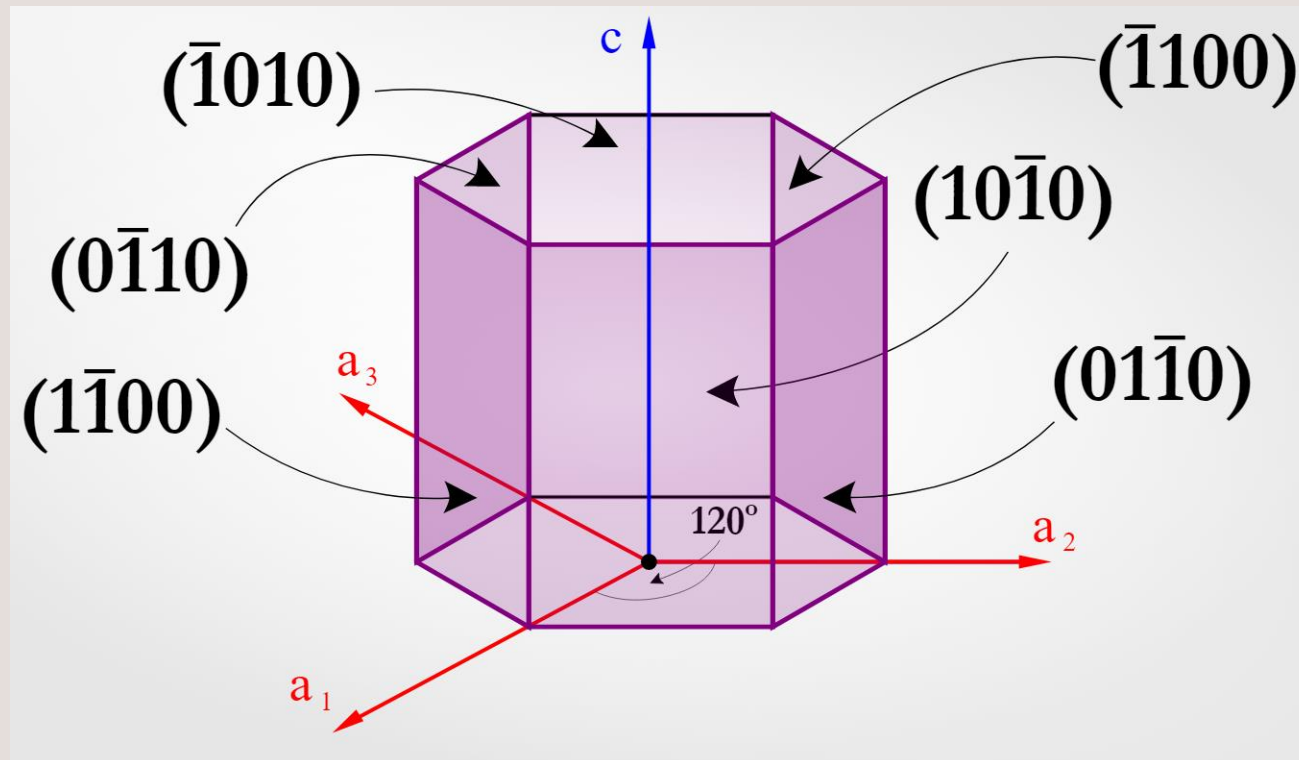
Crystallographic planes and directions in Hexagonal unit cells

Another way to think about it, is to treat the a_3 axis just like the other axes. For example, the $(11\bar{2}0)$ intersects the a_1 axis at 1, the a_2 axis at 1, the a_3 axis at $-\frac{1}{2}$, and the c axis at infinity (never, because it's parallel).



Crystallographic planes and directions in Hexagonal unit cells

The indices of six faces now become $(10\bar{1}0)$, $(01\bar{1}0)$, $(\bar{1}100)$, $(\bar{1}010)$, $(0\bar{1}10)$, $(1\bar{1}00)$ which are now equivalent and belong to the $\{10\bar{1}0\}$ family of planes.



Crystallographic planes and directions in Hexagonal unit cells

Directions in a Hexagonal cell

- Like planes, directions in the hexagonal system are also written in terms of four indices as $[wxyz]$.
- If $[XYZ]$ are index in 3-axis then it can be converted to 4-axis index $[wxyz]$ using the following relations.

$$X = w - y$$

$$Y = x - y$$

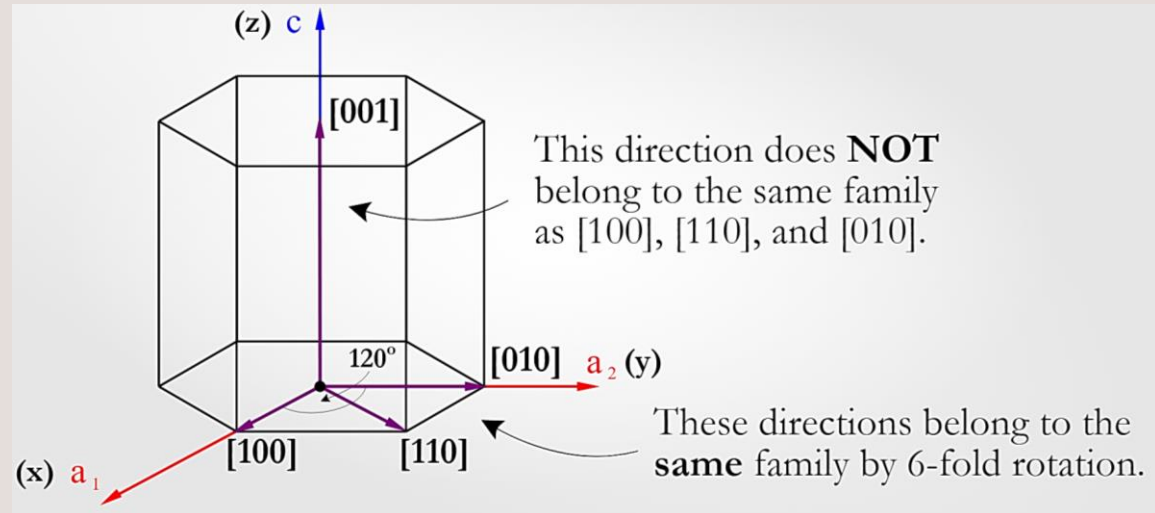
$$Z = z$$

$$w = (2X - Y)/3$$

$$x = (2Y - X)/3$$

$$y = -(X + Y)/3$$

$$z = Z$$



$$\text{Ex: } [100] = [2 \ \bar{1} \ \bar{1} \ 0], \ [210] = [1 \ 0 \ \bar{1} \ 0]$$

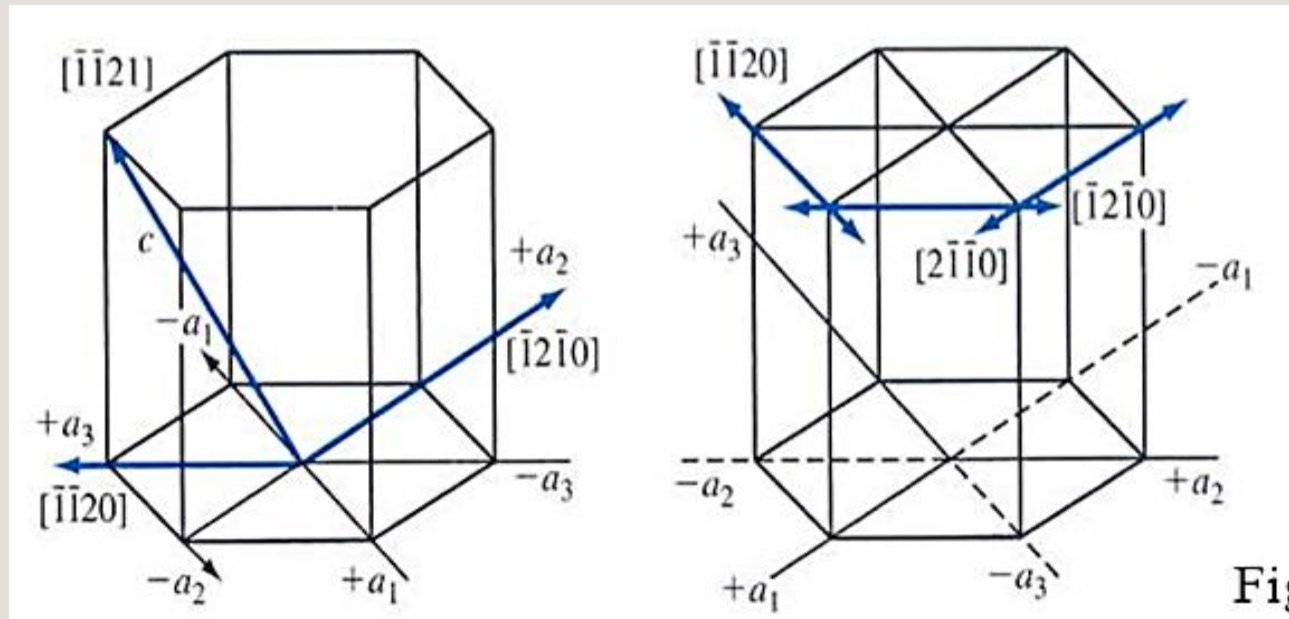
Crystallographic planes and directions in Hexagonal unit cells

Directions in a Hexagonal cell

- Indicated by 4 indices $[uvw]$
- u , v , t , and w are lattice vectors in a_1 , a_2 , a_3 and c directions

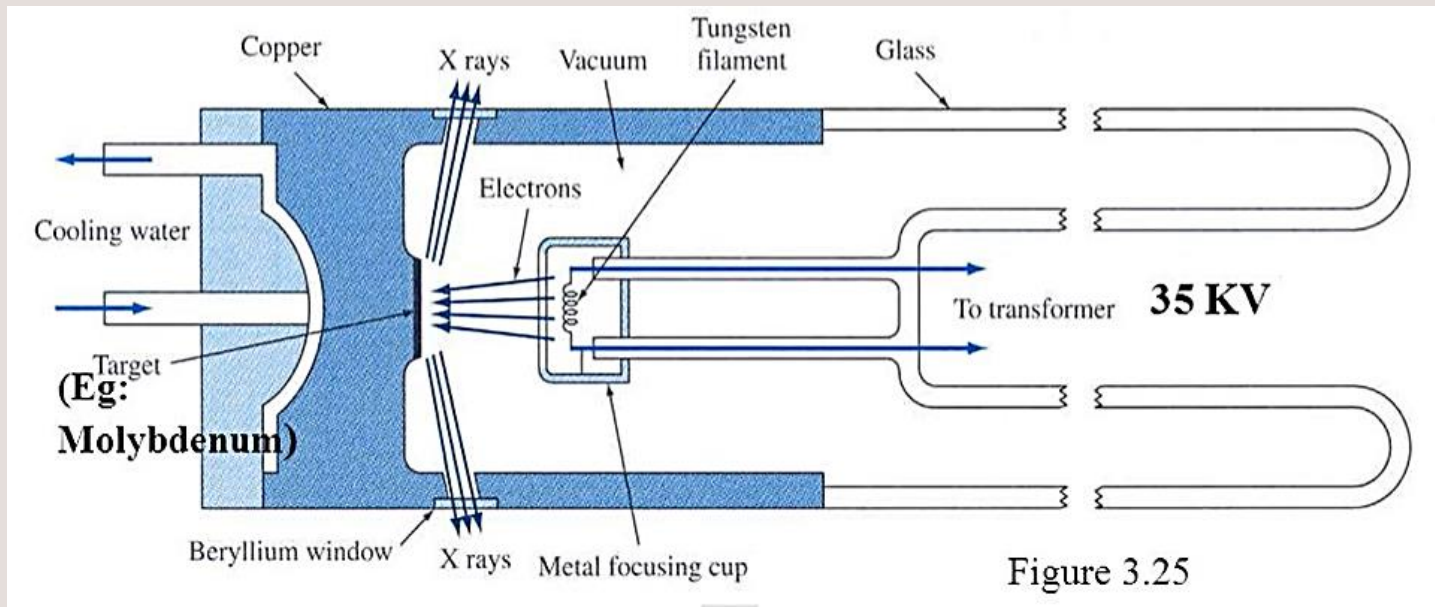
Example:

For a_1 , a_2 , a_3 directions, the direction indices are $[2\bar{1}10]$, $[\bar{1}2\bar{1}0]$ and $[\bar{1}\bar{1}20]$



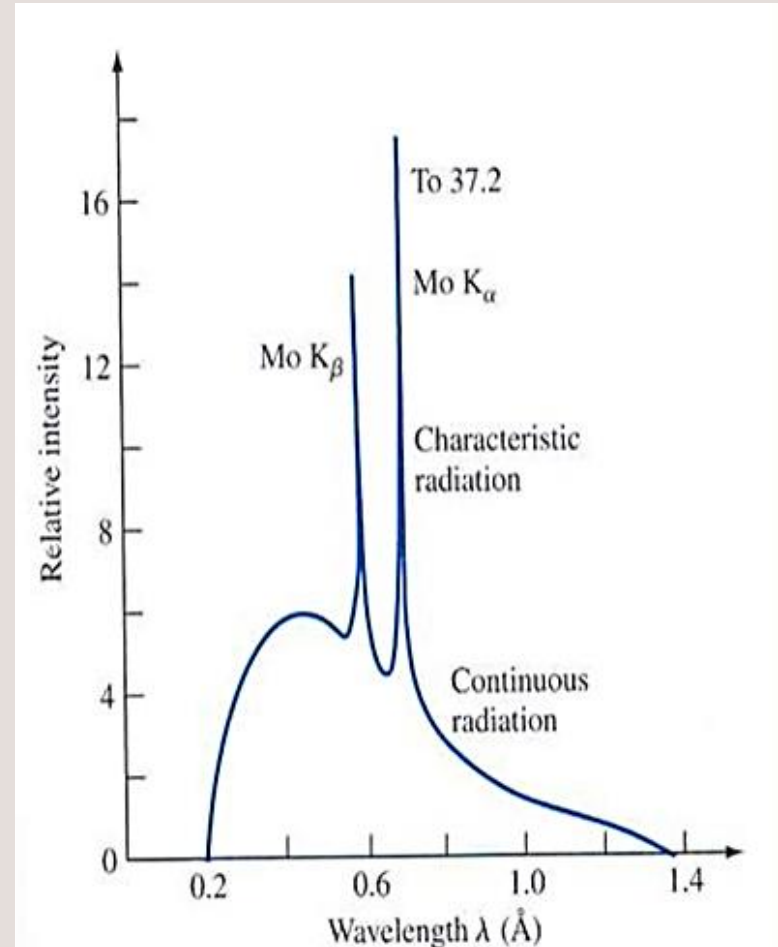
Crystal Structure Analysis

- Information about crystal structure are obtained using X-rays.
- The X-rays used are about the same wavelength (0.05-0.25 nm) or (0.5-2.5 Å) as distance between crystal lattice planes.



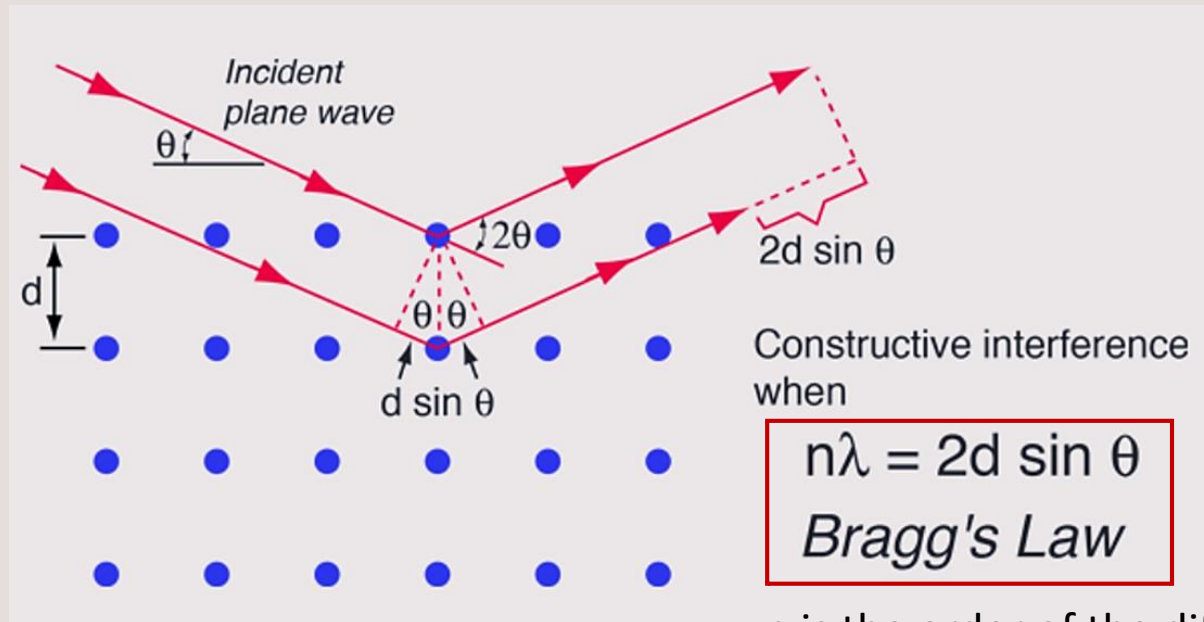
X-Ray Spectrum of Molybdenum

- X-Ray spectrum of Molybdenum is obtained when Molybdenum is used as **target metal**.
- K_α and K_β are characteristic of an element.
- For Molybdenum K_α occurs at wavelength of about 0.07nm (0.7 Å).
- Electrons of $n=1$ shell of target metal are knocked out by bombarding electrons.
- Electrons of higher level drop down by releasing energy to replace lost electrons.



X-Ray Diffraction

- Crystal planes of target metal act as **mirrors** reflecting X-ray beam.
- For rays reflected from different planes to be in phase, the extra distance traveled by a ray should be an integral multiple of wave length λ .



n is the order of the diffraction

Interpreting Diffraction Data

We know that

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

Since

$$\lambda = 2d \sin \theta$$

Substituting for d,

$$\lambda = \frac{2a \sin \theta}{\sqrt{h^2 + k^2 + l^2}}$$

Therefore

$$\sin^2 \theta = \frac{\lambda^2 \sqrt{h^2 + k^2 + l^2}}{4a^2}$$

Note that the wavelength λ and lattice constant a are the same
For both incoming and outgoing radiation.

Interpreting Diffraction Data

For planes 'A' and 'B' we get two equations

$$\sin^2 \theta_A = \frac{\lambda^2 (h_A^2 + k_A^2 + l_A^2)}{4a^2} \quad (\text{For plane 'A'})$$

$$\sin^2 \theta_B = \frac{\lambda^2 (h_B^2 + k_B^2 + l_B^2)}{4a^2} \quad (\text{For plane 'B'})$$

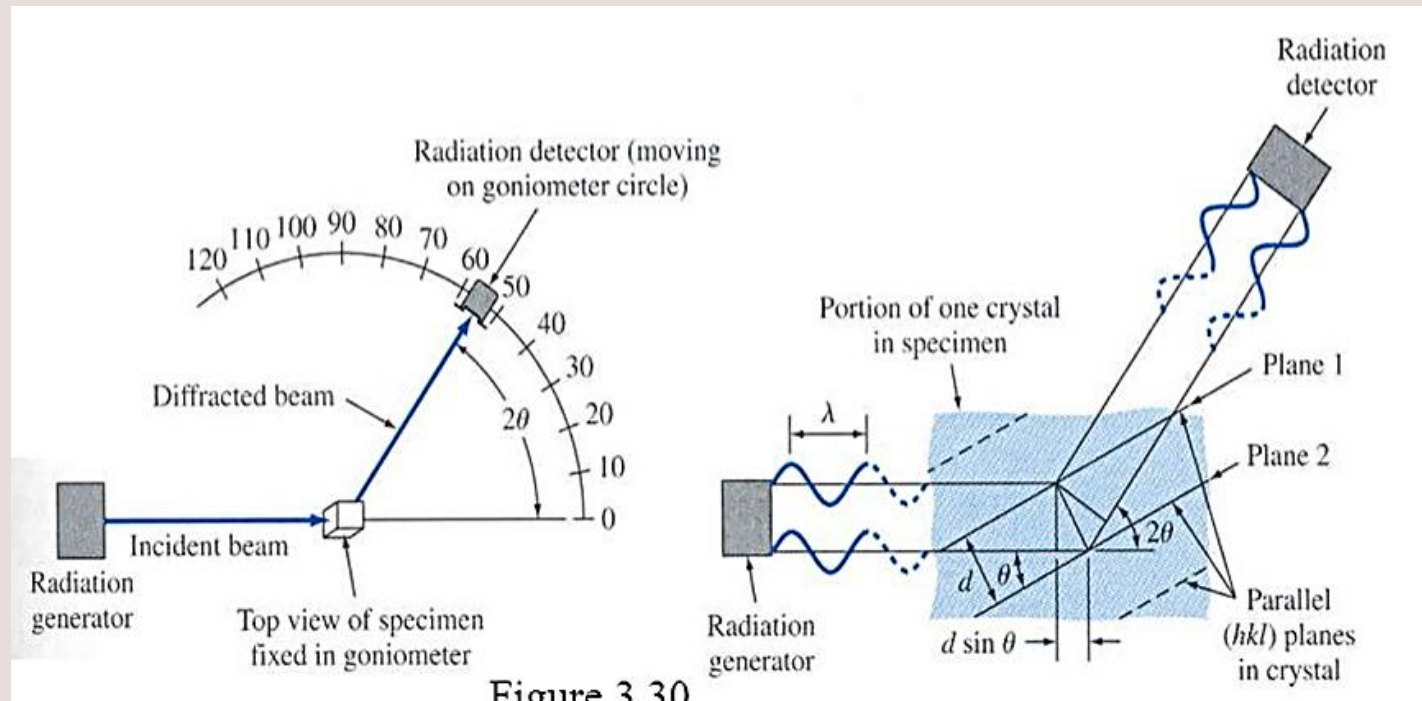
Dividing each other, we get

$$\frac{\sin^2 \theta_A}{\sin^2 \theta_B} = \frac{(h_A^2 + k_A^2 + l_A^2)}{(h_B^2 + k_B^2 + l_B^2)}$$

When θ_A and θ_B is angle of X-ray diffraction from $\{h_A k_A l_A\}$ and $\{h_B k_B l_B\}$ planes.

X-Ray Diffraction Analysis

- Powdered specimen is used for X-ray diffraction analysis as the random orientation facilitates different angle of incidence.
- Radiation counter detects angle and intensity of diffracted beam.



Interpreting Experimental Data

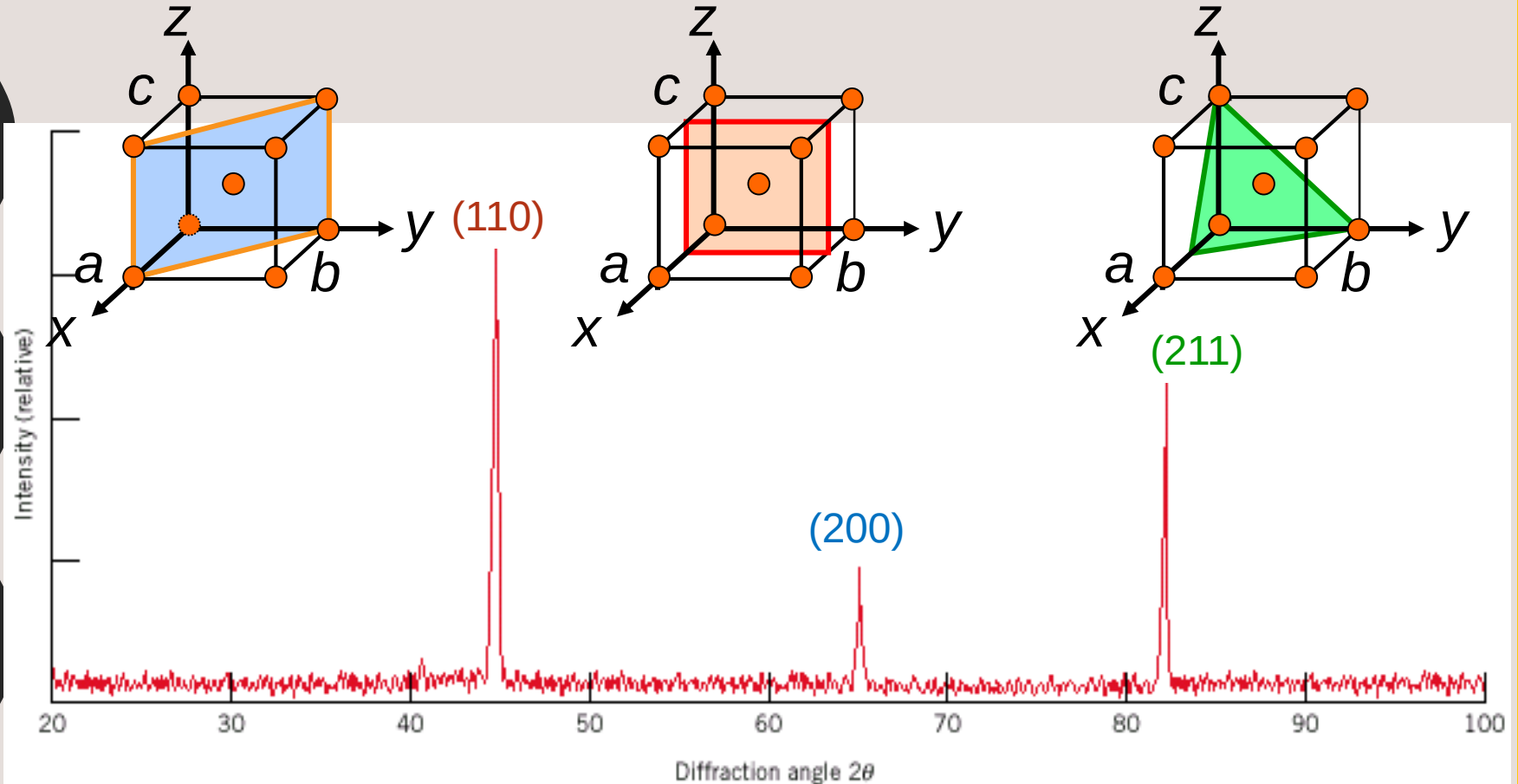
- For BCC crystals, the first two sets of diffracting planes are {110} and {200} planes.

Therefore
$$\frac{\sin^2 \theta_A}{\sin^2 \theta_B} = \frac{(1^2 + 1^2 + 0^2)}{(2^2 + 0^2 + 0^2)} = 0.5$$

- For FCC crystals the first two sets of diffracting planes are {111} and {200} planes

Therefore
$$\frac{\sin^2 \theta_A}{\sin^2 \theta_B} = \frac{(1^2 + 1^2 + 1^2)}{(2^2 + 0^2 + 0^2)} = 0.75$$

X-Ray Diffraction Pattern



Diffraction pattern for polycrystalline α -iron (BCC)